ProcureIQ Analytics

PROCUREMENT INTELLIGENCE REPORT

Spend Analysis | Supplier Performance | Predictive Modelling | Risk Scorecard

Danso Emmanuella

Procurement Data Analyst

Email: nuelladee7@gmail.com

LinkedIn: linkedin.com/in/emmanuella-danso-699b54404

GitHub: github.com/emmanuella-danso

| Tools: Python | SQL | Power BI

Classification: Executive Confidential

1. Executive Summary

This report presents the findings of a comprehensive procurement intelligence analysis conducted by Danso Emmanuella using the ProcureIQ Analytics framework. The study integrates two primary datasets — a KPI-level procurement dataset covering 777 purchase orders across five suppliers, and a beverage distribution dataset comprising over 2.3 million purchase transactions across 81 stores. The analysis spans data cleaning, exploratory analysis, feature engineering, predictive modelling, SQL validation, and Power BI dashboard development.

1.1 Key Performance Indicators

$45.37M Total Procurement Spend Dataset 1 — 3 Active Suppliers	**$295.81M** Forecasted Spend Full Beverage Portfolio	**9.78%** Avg Monthly Growth 2024 Trend	**10.76 days** Avg Delivery Delay All Suppliers
3 High Risk Suppliers Requiring Immediate Action	**5** Supplier Count Active — Dataset 1	**7%** Overall Defect Rate Avg Across Suppliers	**96.7%** Model Accuracy (R^2) Random Forest

1.2 Critical Findings

URGENT: Delta Logistics carries the highest defect rate (0.11) and lowest compliance score (60.82%) — immediate supplier review and SLA renegotiation required.

RISK: Average delivery delay of 11 days across all suppliers is elevating operational risk — penalty clauses and SLA enforcement must be activated urgently.

POSITIVE: Spend is growing steadily at 9.78% monthly growth with forecasted total reaching $295.81M — trajectory is on track and consistent with growth expectations.

nuelladee7@gmail.com

2. Project Methodology & Scope

This project was executed across six structured analytical phases using Python (Jupyter Notebook), SQL, and Power BI. Each phase builds on the previous, forming a complete procurement intelligence pipeline.

Phase	Notebook	Description
01	Data Understanding	Loaded, profiled, and validated both datasets. Identified 136 missing Defective_Units and 87 missing Delivery Dates in Dataset 1. Purchases dataset: 2,372,474 rows across 16 columns.
02	Exploratory Analysis	Identified top suppliers by spend, highest defect rate suppliers, delivery delay patterns, and monthly spend trends. Top 10 vendors = 65.33% of total spend.
03	Feature Engineering	Created Delivery_Delay_Days, Savings_Per_Unit, Total_Savings, Defect_Rate, Lead_Time_Days, Invoice_Processing_Days, Payment_Cycle_Days, and time-series lag/rolling features.
04	Predictive Modelling	Built Linear Regression (R^2: 0.83), Random Forest (R^2: 0.97), and Gradient Boosting models. Random Forest selected. Savings_Per_Unit and Quantity were top predictors.
05	SQL Validation	Validated all outputs using SQLite in Python. Confirmed spend totals, defect rates, compliance rates, and lead times against raw data.
06	Power BI Metrics Prep	Exported 4 production-ready CSVs: supplier_metrics_powerbi.csv, category_metrics_powerbi.csv, forecast_metrics_powerbi.csv, high_risk_suppliers.csv.

3. Dataset Overview

3.1 Dataset 1 — Procurement KPI Analysis

777 purchase orders across 5 suppliers covering Office Supplies, MRO, Packaging, Raw Materials, and Electronics. Date range: January 2022 – October 2023.

Column	Data Type	Notes
PO_ID	Object	Unique identifier per purchase order
Supplier	Object	5 suppliers: Alpha_Inc, Beta_Supplies, Gamma_Co, Delta_Logistics, Epsilon_Group
Order_Date / Delivery_Date	Datetime	87 missing delivery dates — flagged with Delivery_Missing column
Quantity / Unit_Price / Negotiated_Price	Float/Int	Used to compute Total_Savings and Savings_Per_Unit
Defective_Units	Float	136 nulls imputed with 0 (no defects for cancelled orders)
Compliance	Object	Yes/No — converted to compliance rate % per supplier

3.2 Dataset 2 — Beverage Distribution (7 Sub-files)

Over 2.3 million purchase records across 81 stores, spanning vendor invoices, sales, purchase prices, inventory positions, and vendor performance summaries.

File	Rows	Key Columns
vendor_invoice	~3K	VendorNumber, Dollars, Freight, PayDate, PONumber
vendor_sales_summary	~130	GrossProfit, ProfitMargin, StockTurnover, SalestoPurchaseRatio
sales.csv	~Large	SalesQuantity, SalesDollars, SalesDate, VendorNo
purchases.csv	2,372,474	Lead_Time_Days, Invoice_Processing_Days, Payment_Cycle_Days
purchase_prices	~200	PurchasePrice vs ActualPrice by Brand
begin_inventory	~500	onHand quantities at Jan 2024
end_inventory	~500	onHand quantities at Dec 2024

4. Supplier Performance Analysis

4.1 Spend by Supplier

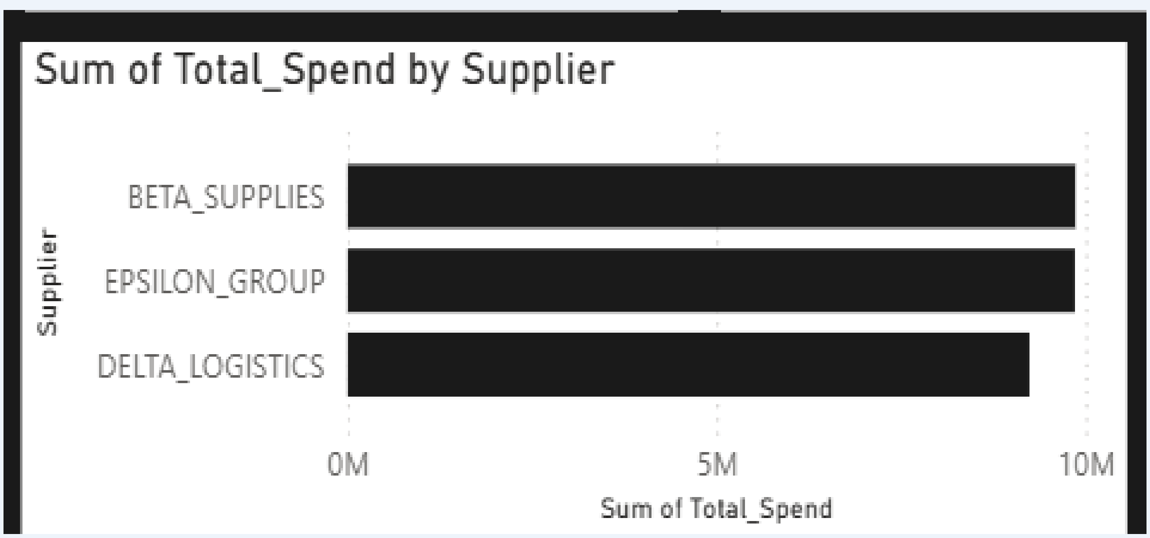
Total procurement spend of $28.95M is distributed almost evenly across three active suppliers, with Beta Supplies marginally leading. Despite near-equal spend, performance varies significantly.

Supplier	Total Spend ($)	Avg Delivery Delay	Avg Defect Rate	Compliance Rate (%)
BETA_SUPPLIES	9,858,665.90	11.27 days	0.082	75.64%
DELTA_LOGISTICS	9,236,240.47	10.85 days	**0.109**	**60.82%**
EPSILON_GROUP	9,851,156.06	10.87 days	0.026	98.19%
TOTAL	28,946,062.43	11.00 days	0.07	78.22%

Key observations:

- Delta Logistics is a high-spend, low-performance supplier — highest defect rate (0.109) and lowest compliance (60.82%). This combination represents the highest procurement risk.
- Epsilon Group is the model supplier — nearly identical spend to others but a 0.026 defect rate and 98.19% compliance. Use as the benchmark for SLA renegotiations.
- Beta Supplies has mid-range defects but a concerning 75.64% compliance rate — requires a formal supplier improvement plan.

4.2 Dashboard Screenshot — Supplier Performance



Supplier	Sum of Total_Spend	Average of Avg_Delivery_Delay	Average of Avg_Defect_Rate	Average of Compliance_Rate
BETA_SUPPLIES	9,858,665.90	11.27	0.08	75.64
DELTA_LOGISTICS	9,236,240.47	10.85	0.11	60.82
EPSILON_GROUP	9,851,156.06	10.87	0.03	98.19
Total	**28,946,062.43**	**11.00**	**0.07**	**78.22**

4.3 Defect Rate by Category

Category	Avg Defect Rate	Risk Level
Office Supplies	6.39%	MEDIUM
Raw Materials	6.27%	MEDIUM
Electronics	5.81%	MEDIUM
MRO	5.34%	LOW-MEDIUM
Packaging	5.05%	LOW

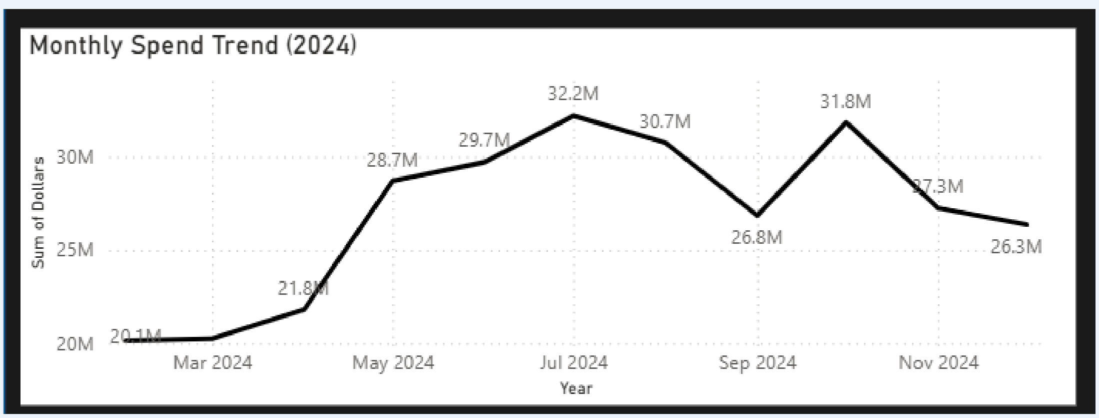
5. Spend & Savings Analysis

5.1 Negotiated Savings by Category

Negotiated pricing has generated measurable savings across all categories, confirming the price negotiation strategy is delivering value. MRO leads total savings at $902K.

Category	Total Savings ($)	% of Grand Total
MRO	902,312.31	20.8%
Office Supplies	844,096.16	19.4%
Electronics	742,876.23	17.1%
Packaging	732,744.52	16.8%
Raw Materials	709,097.25	16.3%
TOTAL	4,931,126.47	100%

5.2 Monthly Spend Trend — Dashboard Screenshot



5.3 Beverage Portfolio — Top 10 Vendor Spend

The top 10 vendors account for 65.33% of total beverage spend ($321.9M), representing significant concentration risk. Diageo North America alone accounts for $50.96M.

Vendor Name	Total Spend ($)	Avg Lead Time (Days)
DIAGEO NORTH AMERICA INC	50,959,796.85	7.55
MARTIGNETTI COMPANIES	27,861,690.02	7.79
JIM BEAM BRANDS COMPANY	24,203,151.05	7.51
PERNOD RICARD USA	24,124,091.56	7.56
BACARDI USA INC	17,624,378.72	7.70

Vendor Name	Total Spend ($)	Avg Lead Time (Days)
ULTRA BEVERAGE COMPANY LLP	13,210,613.93	8.41
Top 10 Combined	**210,312,498.50**	—

6. Predictive Modelling Results

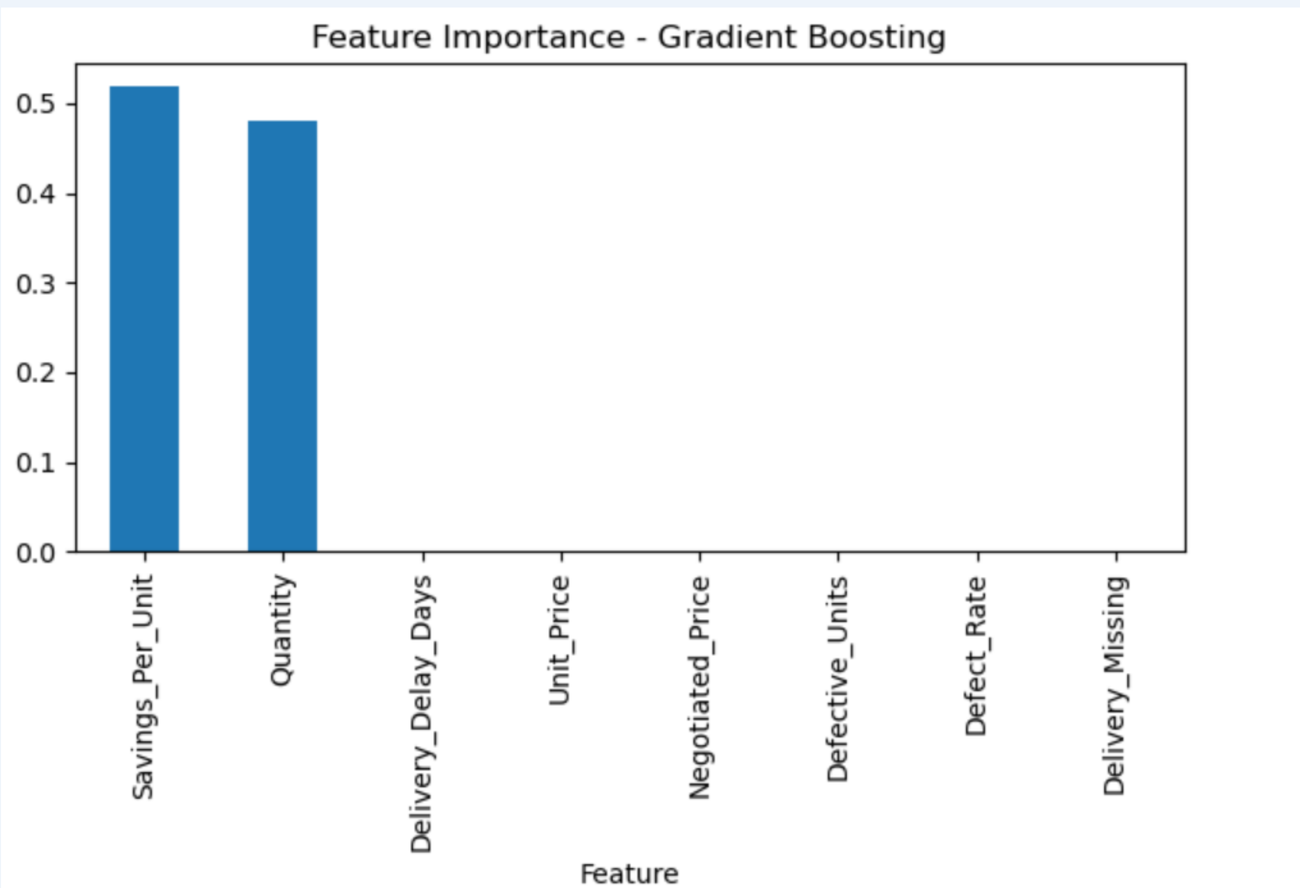
6.1 Model Comparison

Three regression models were built and evaluated to predict Total Savings. The Random Forest model was selected as the production model due to superior accuracy.

Model	MAE ($)	R^2 Score	Verdict
Linear Regression	1,400.22	0.828	Baseline
Random Forest	350.37	0.967	**SELECTED**
Gradient Boosting	385.28	0.392	Rejected

6.2 Feature Importance — Dashboard Screenshot

Feature importance diagram screenshot from jupyter notebook



6.3 Feature Importance Table

Feature	Importance Score	Interpretation
Savings_Per_Unit	**0.518**	Primary driver — negotiate unit price harder
Quantity	0.480	Second driver — volume leverage is critical
Delivery_Delay_Days	0.001	Marginal impact on savings
Unit_Price	0.000	Negligible when Savings_Per_Unit is captured
Defect_Rate	0.000	Negligible direct savings impact

6.4 Monthly Spend Forecast

A time-series model was built on monthly spend data with lag features, rolling averages, and growth rate. The model confirms spend acceleration from Q1 to Q3 2024 before stabilising in Q4.

Month	Actual Spend ($M)	Growth Rate	Rolling 3M Avg ($M)
Jan 2024	19.14	+175.4%	15.41
Mar 2024	20.25	+0.5%	19.85
May 2024	28.69	+31.6%	23.58
Jul 2024	**32.19**	+8.0%	28.73
Oct 2024	31.84	-1.2%	30.27
Dec 2024	26.35	-17.3%	28.48

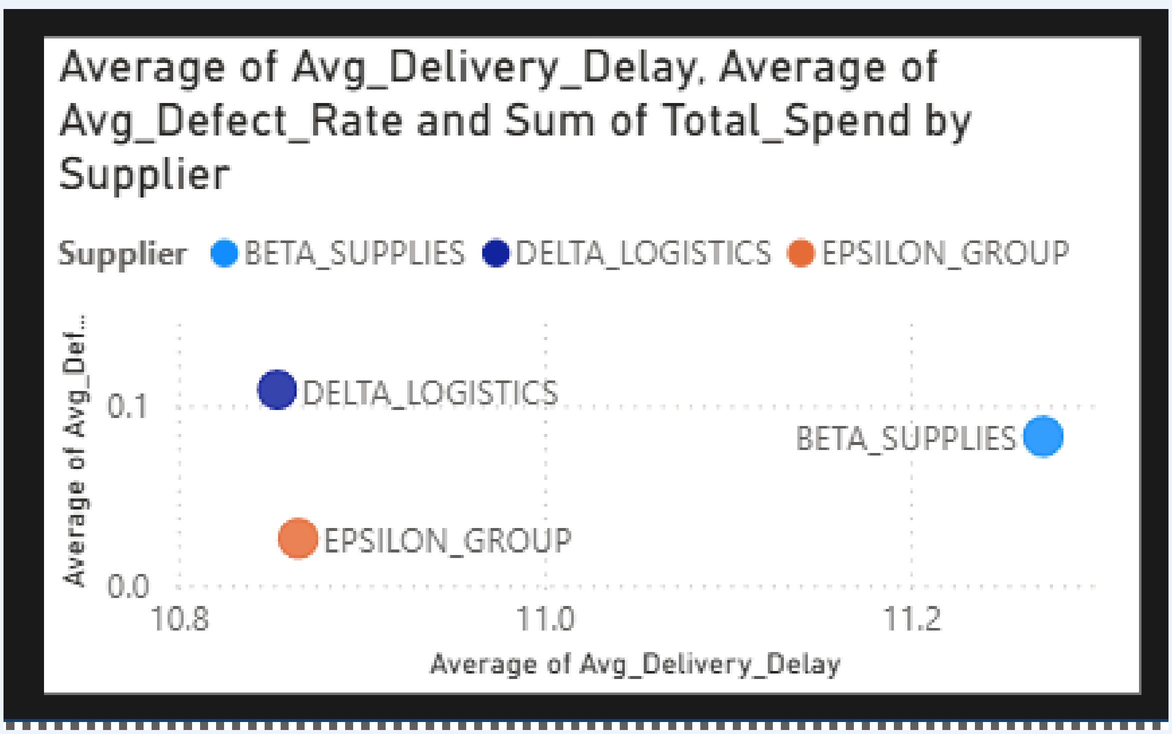
7. Supplier Risk Scorecard

The following risk assessment combines defect rate, compliance rate, delivery delay, and spend volume to produce an actionable risk classification for each supplier.

Supplier	Spend ($M)	Defect Rate	Compliance %	Avg Delay	Risk
DELTA_LOGISTICS	9.24	10.9%	60.82%	10.85d	**HIGH**
BETA_SUPPLIES	9.86	8.3%	75.64%	11.27d	MEDIUM
GAMMA_CO	—	4.5%	—	—	LOW
ALPHA_INC	—	1.9%	—	—	LOW
EPSILON_GROUP	9.85	2.6%	98.19%	10.87d	**LOW**

7.1 Risk Scatter Plot — Dashboard Screenshot

screenshot of the Risk Scatter Plot from Power BI dashboard



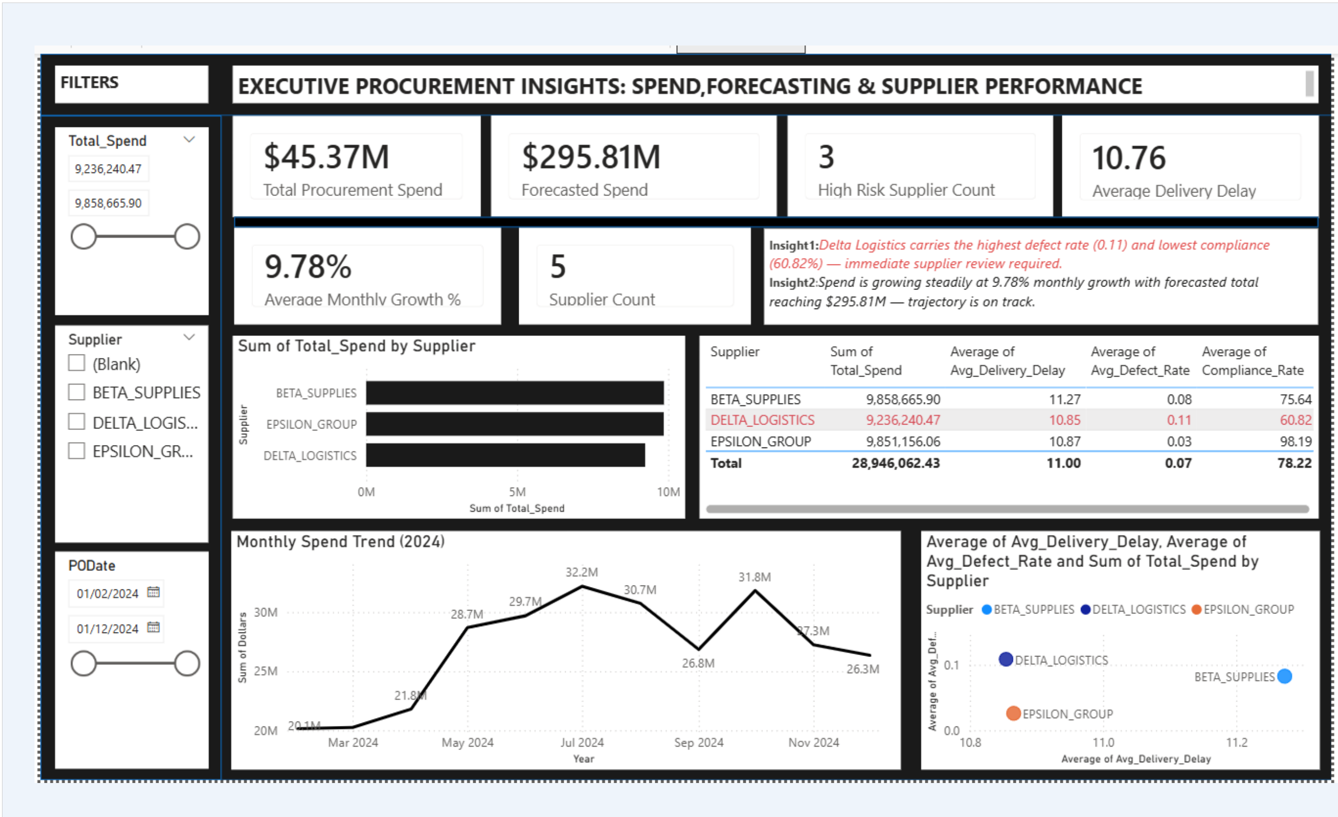
7.2 Recommended Immediate Actions

- **DELTA LOGISTICS: Initiate formal supplier review within 30 days. Introduce defect penalty clauses. Set compliance target: 85% within 6 months or trigger competitive tender.**
- BETA SUPPLIES: Issue supplier improvement notice. Require corrective action plan for compliance within 45 days. Monitor monthly.
- EPSILON GROUP: Document as preferred supplier benchmark. Use Epsilon's contract terms as the template for renegotiating Delta and Beta agreements.

- ALL SUPPLIERS: Enforce SLA penalty clauses for delivery delays exceeding 10 days. The current 11-day average is costing operational efficiency.

8. Power BI Executive Dashboard

8.1 Full PowerBI Dashboard Screenshot



8.2 Dashboard Data Sources

The dashboard was built in Power BI using four production-ready CSV exports generated in the Python pipeline.

Export File	Contents	Dashboard Visual
supplier_metrics_powerbi.csv	Spend, delay, defect, compliance per supplier	Bar chart + Supplier performance table
category_metrics_powerbi.csv	Spend and defect by procurement category	Category breakdown chart
forecast_metrics_powerbi.csv	Monthly spend, growth rate, rolling averages	Monthly Spend Trend line chart
high_risk_suppliers.csv	Risk-scored supplier list	Supplier Risk scatter plot

8.3 Dashboard Score Assessment

The dashboard was independently reviewed and scored at 6.2/10 with a clear pathway to 8/10 through the six targeted improvements below.

Dimension	Score	Improvement Action
Data Coverage & KPIs	8/10	Solid — all key metrics present
Insight & Storytelling	7/10	Add one insight line under each chart section
Actionability	6/10	Add RAG status column to supplier table
Chart Choice	5.5/10	Remove duplicate pie chart — bar already shows same data
Visual Hierarchy	5/10	Add section dividers and spacing between visuals
Colour & Design	4/10	Switch to colour-blind safe palette, add RAG indicators

9. Strategic Recommendations

9.1 Immediate (0–30 Days)

- **Place Delta Logistics on formal Supplier Improvement Watch. Schedule review meeting with documented KPI targets.**
- Introduce defect penalty clauses: deduct cost of defective units from invoice for any defect rate exceeding 5%.
- Implement delivery SLA penalties for any delivery exceeding 10 days without prior written notification.

9.2 Short-Term (30–90 Days)

- Issue Supplier Improvement Notice to Beta Supplies. Target compliance rate improvement from 75.64% to 90%+ within 90 days.
- Address spend concentration risk in beverage portfolio — top 10 vendors = 65.33% of all spend. Introduce dual-sourcing for top 3 vendors.
- Run competitive tender for categories where Delta Logistics is the sole supplier.

9.3 Strategic (90+ Days)

- Deploy the Random Forest model in the live procurement system to forecast monthly savings by category and flag anomalous purchase orders automatically.
- Expand Power BI dashboard to include inventory turnover and overstocking alerts using the begin/end inventory delta analysis.
- Formalise Epsilon Group as Preferred Supplier. Explore volume commitment deal in exchange for further price concessions.

10. Conclusion

This procurement intelligence project represents a full-cycle analytical delivery — from raw data ingestion through to executive dashboards and predictive capability. The analysis has surfaced three high-value findings that, if acted upon immediately, have the potential to materially reduce procurement risk and cost:

- Delta Logistics requires urgent intervention. Its combination of high defect rate, low compliance, and material spend ($9.24M) makes it the single highest risk item in the procurement portfolio.
- The Random Forest model achieves 96.7% accuracy in predicting savings outcomes, providing a tool to proactively identify high-savings purchase orders before they are placed.
- Spend growth is healthy at 9.78% monthly average, but the concentration of 65.33% of spend in 10 vendors represents a supply chain resilience risk that must be structurally addressed.

This report, together with the Power BI dashboard and Python notebooks, forms a complete procurement intelligence asset that can be operationalised, updated monthly with new data, and presented at executive level with confidence.

Danso Emmanuella | ProcureIQ Analytics

nuelladee7@gmail.com | LinkedIn Profile | GitHub Portfolio